

V3 empirical study · Instruction

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based models

What this report is

- 1.5-year lookback, monthly rolling, 10,000 LSM paths. Models: MD-GBM, GBM, GARCH, Merton, and GARCH-Merton. P dynamics come from lookback returns. GBM / MD-GBM use an additive Q shift; GARCH uses Duan LRNV; Merton uses Pan jump-size premium μ_J^* from listed calls; GARCH-Merton uses Duan LRNV plus Pan μ_J^* . MD-GBM is Markov Directional Geometric Brownian Motion: 1-lag Modified GBM with folded-normal means matched to sample $|R|$.
- Companies are SPY, AAPL, MSFT, and AMZN.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 5 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Ranking column ranks 1 = best.

Experimental design (held fixed for every model)

- Models: MD-GBM (Markov Directional Geometric Brownian Motion), GBM, GARCH(1,1), Merton jump-diffusion, GARCH-Merton.
- MD-GBM means Markov Directional Geometric Brownian Motion. In this study it is specified in V4_MD-GBM_model.pdf in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based models

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all five models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to the LSM comparison set: (1) calls only, finite S, K, C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) near ATM $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- P dynamics are estimated from lookback log returns. Merton / GARCH-Merton jump-size premia for Q still use listed calls (Pan). LSM prices the same filtered Monday ATM sample for every model.

Model-correctness adjustments (V3)

- GARCH uses Duan LRNVR. Merton uses Pan μ_J^* from listed calls. GBM / MD-GBM use an additive shift so $E[e^R] = e^{r_f \Delta t}$.
- LSM: Longstaff-Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE\%} = 100 \times \text{mean}(|C_{\text{model}} - C_{\text{mkt}}| / C_{\text{mkt}})$. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})$. Positive bias = model expensive vs market.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all five models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date $\leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All five models in this report still share that exact sample.

GBM / MD-GBM / GARCH estimate from lookback log returns only. Merton / GARCH–Merton add 3σ residual jumps; Q jump-size premia use listed calls.

Page 4 · Table block · SPY · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	41.02	32.65	-4.71	1
GBM	55.01	40.97	+21.16	5
GARCH	43.04	35.72	+33.41	2
Merton	51.73	37.42	+25.72	4
GARCH-Merton	44.31	37.49	+36.30	3

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	29.99	24.17	+13.83	1
GBM	38.08	30.45	+24.88	5
GARCH	36.35	29.04	+22.86	2
Merton	37.32	29.83	+23.72	3
GARCH-Merton	37.90	30.58	+24.87	4

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	36.48	31.07	+14.03	1
GBM	49.51	41.57	+32.90	2
GARCH	65.98	49.13	+41.26	4
Merton	60.60	52.40	+46.39	3
GARCH-Merton	68.12	51.50	+44.06	5

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	43.63	36.23	+14.85	1
GBM	64.88	57.43	+44.13	4
GARCH	53.62	42.08	+35.24	3
Merton	70.92	64.09	+53.08	5
GARCH-Merton	52.76	41.90	+35.13	2

Page 5 · Table block · Company summary · five models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookback, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	37.97	29.51	+5.30	2
GBM	49.54	37.95	+23.71	5
GARCH	37.65	31.10	+23.97	1
Merton	47.89	36.89	+30.25	4
GARCH-Merton	45.49	38.29	+35.63	3

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	30.15	24.58	+14.84	1
GBM	47.93	39.44	+36.05	3
GARCH	53.53	43.66	+39.14	4
Merton	46.87	37.36	+34.12	2
GARCH-Merton	75.13	64.41	+64.13	5

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n} = 51$

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	27.95	24.23	+7.20	1
GBM	39.31	33.51	+23.84	3
GARCH	36.84	29.16	+21.78	2
Merton	43.10	37.05	+32.25	4
GARCH-Merton	49.41	40.98	+37.98	5

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	34.58	27.59	+9.56	1
GBM	48.18	39.74	+30.10	3
GARCH	41.61	31.18	+23.29	2
Merton	52.19	44.42	+37.76	4
GARCH-Merton	52.64	41.77	+36.69	5

RMSE%, MAE%, and Bias% are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMSE% only.

Page 6 · Table block · AAPL · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	43.89	32.72	+12.84	3
GBM	57.67	42.83	+31.10	5
GARCH	34.36	28.77	+25.46	1
Merton	55.37	40.83	+36.40	4
GARCH-Merton	40.33	34.50	+32.14	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	30.36	23.26	+17.13	1
GBM	50.20	39.70	+38.63	4
GARCH	40.08	29.28	+23.86	2
Merton	48.87	38.01	+37.02	3
GARCH-Merton	59.86	50.37	+50.37	5

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	23.64	20.27	+5.37	1
GBM	32.53	28.14	+18.76	2
GARCH	40.82	29.83	+21.75	4
Merton	36.65	31.82	+28.29	3
GARCH-Merton	54.71	44.78	+41.77	5

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	31.97	25.38	+9.95	1
GBM	46.19	39.52	+30.76	3
GARCH	45.53	33.84	+29.35	2
Merton	47.03	40.14	+35.87	4
GARCH-Merton	57.26	46.26	+44.06	5

Page 7 · Table block · MSFT · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	34.43	27.73	+3.32	1
GBM	44.02	34.76	+17.99	3
GARCH	50.93	41.05	+35.87	4
Merton	41.80	34.10	+23.08	2
GARCH-Merton	61.38	51.11	+48.59	5

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	38.79	31.25	+28.02	1
GBM	62.49	53.14	+52.98	3
GARCH	85.94	73.78	+73.69	4
Merton	56.13	47.20	+46.76	2
GARCH-Merton	108.47	97.36	+97.36	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	25.61	21.67	+4.03	1
GBM	41.73	34.38	+27.55	3
GARCH	37.25	30.23	+24.44	2
Merton	43.31	35.57	+33.24	4
GARCH-Merton	48.79	40.81	+38.95	5

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	31.75	23.91	+4.32	1
GBM	45.93	37.46	+28.12	2
GARCH	49.20	34.48	+25.36	3
Merton	50.06	41.56	+34.80	4
GARCH-Merton	54.17	40.27	+34.17	5

Page 8 · Table block · AMZN · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	35.59	28.09	-0.26	3
GBM	46.92	36.28	+22.05	5
GARCH	27.67	23.47	+10.57	1
Merton	46.51	35.73	+31.26	4
GARCH-Merton	34.76	29.27	+26.15	2

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	21.30	19.23	-0.63	1
GBM	31.09	25.49	+16.54	2
GARCH	34.57	27.93	+19.87	3
Merton	35.62	26.86	+18.58	4
GARCH-Merton	57.06	45.50	+44.68	5

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	34.59	30.76	+12.20	2
GBM	43.66	38.02	+25.21	3
GARCH	32.44	27.42	+19.15	1
Merton	49.34	43.76	+35.22	5
GARCH-Merton	44.74	37.34	+33.22	4

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	40.03	33.48	+14.41	2
GBM	52.42	42.24	+31.41	4
GARCH	30.12	25.22	+15.17	1
Merton	59.46	51.55	+42.61	5
GARCH-Merton	46.47	38.78	+31.83	3

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	MD-GBM	33.94	34.51	12
2	GARCH	44.24	40.45	4
3	GBM	47.65	46.56	0
4	Merton	49.42	49.10	0
5	GARCH-Merton	54.44	53.47	0

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
MD-GBM	4	3	4	1	12
GBM	0	0	0	0	0
GARCH	0	1	0	3	4
Merton	0	0	0	0	0
GARCH-Merton	0	0	0	0	0

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
MD-GBM	38.73	30.11	30.08	36.85	33.94
GBM	50.90	45.47	41.86	52.35	47.65
GARCH	39.00	49.23	44.12	44.61	44.24
Merton	48.85	44.48	47.47	56.87	49.42
GARCH-Merton	45.20	65.82	54.09	52.67	54.44

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
MD-GBM	37.78	38.75	4
GBM	51.87	52.26	0
GARCH	49.75	48.33	0
Merton	55.14	56.16	0
GARCH-Merton	50.77	48.54	0

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
MD-GBM	32.47	31.17	3
GBM	46.65	48.20	0
GARCH	40.20	40.45	1
Merton	46.98	47.95	0
GARCH-Merton	53.04	55.99	0

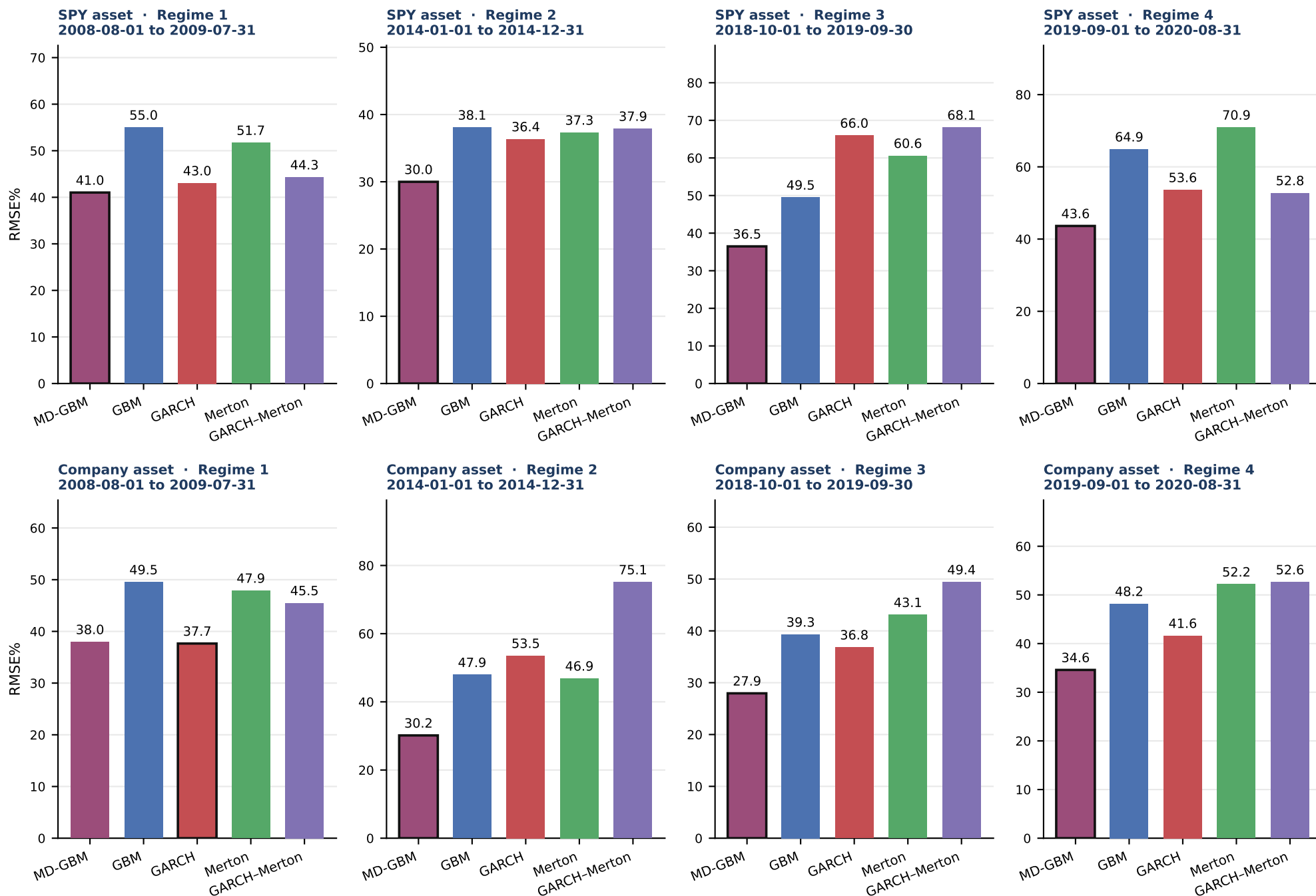
Table S4.3 · MSFT

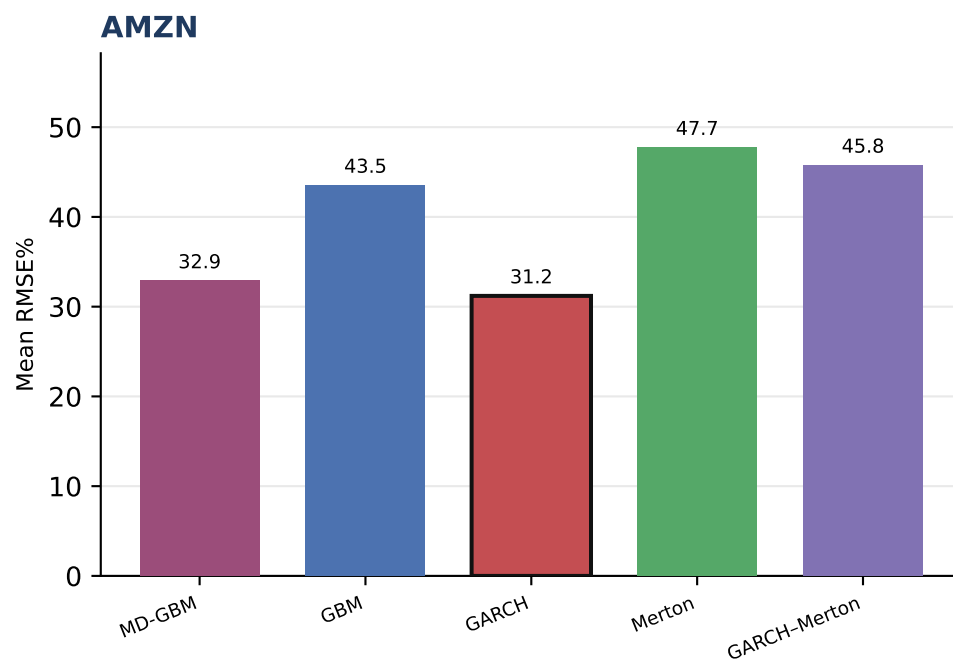
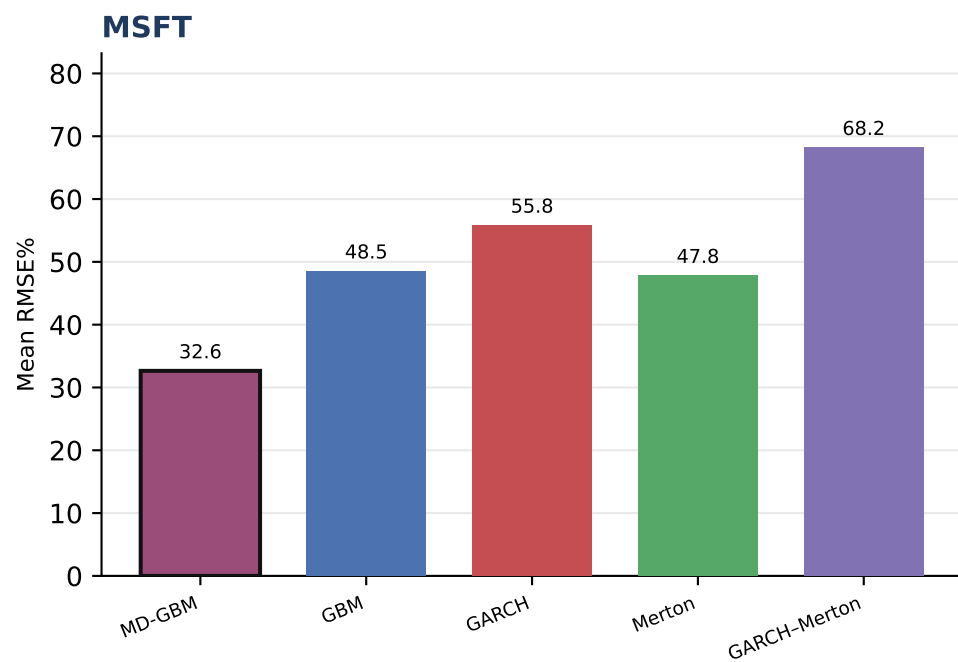
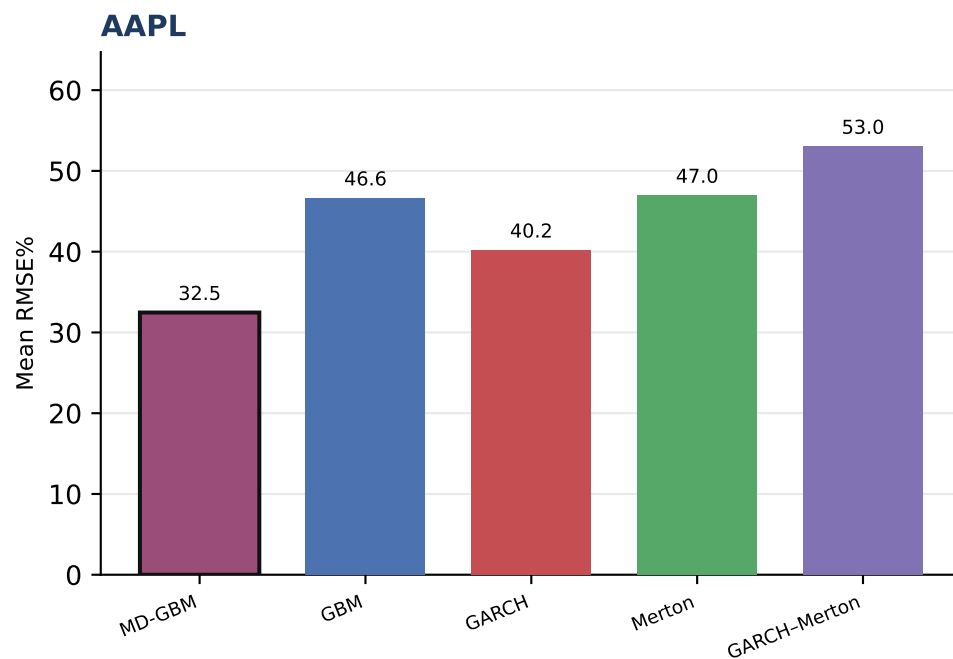
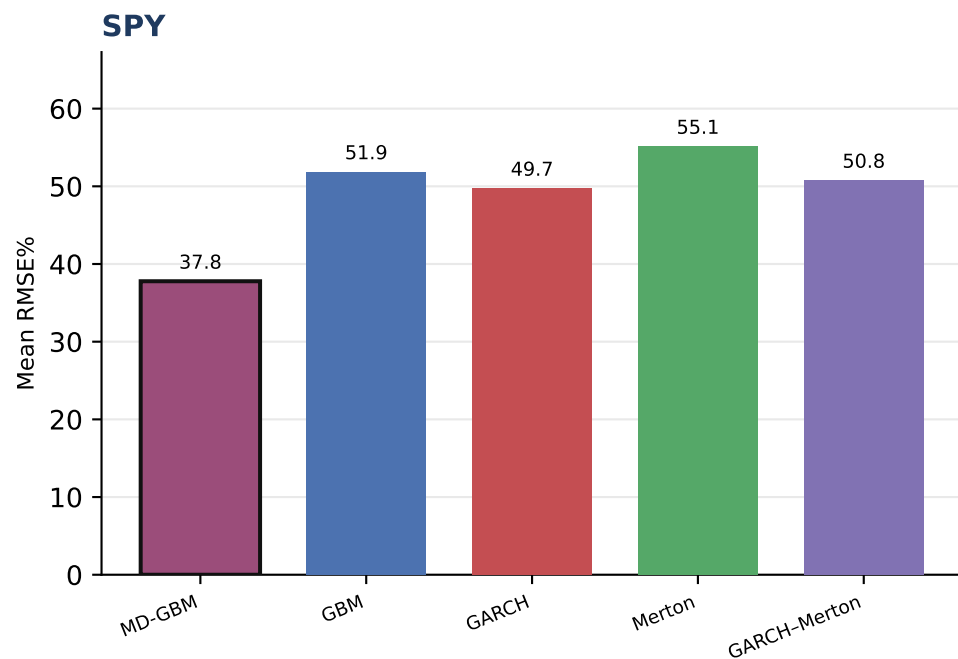
Model	Mean RMSE%	Median	# regimes best
MD-GBM	32.65	33.09	4
GBM	48.54	44.98	0
GARCH	55.83	50.06	0
Merton	47.83	46.69	0
GARCH-Merton	68.20	57.78	0

Table S4.4 · AMZN

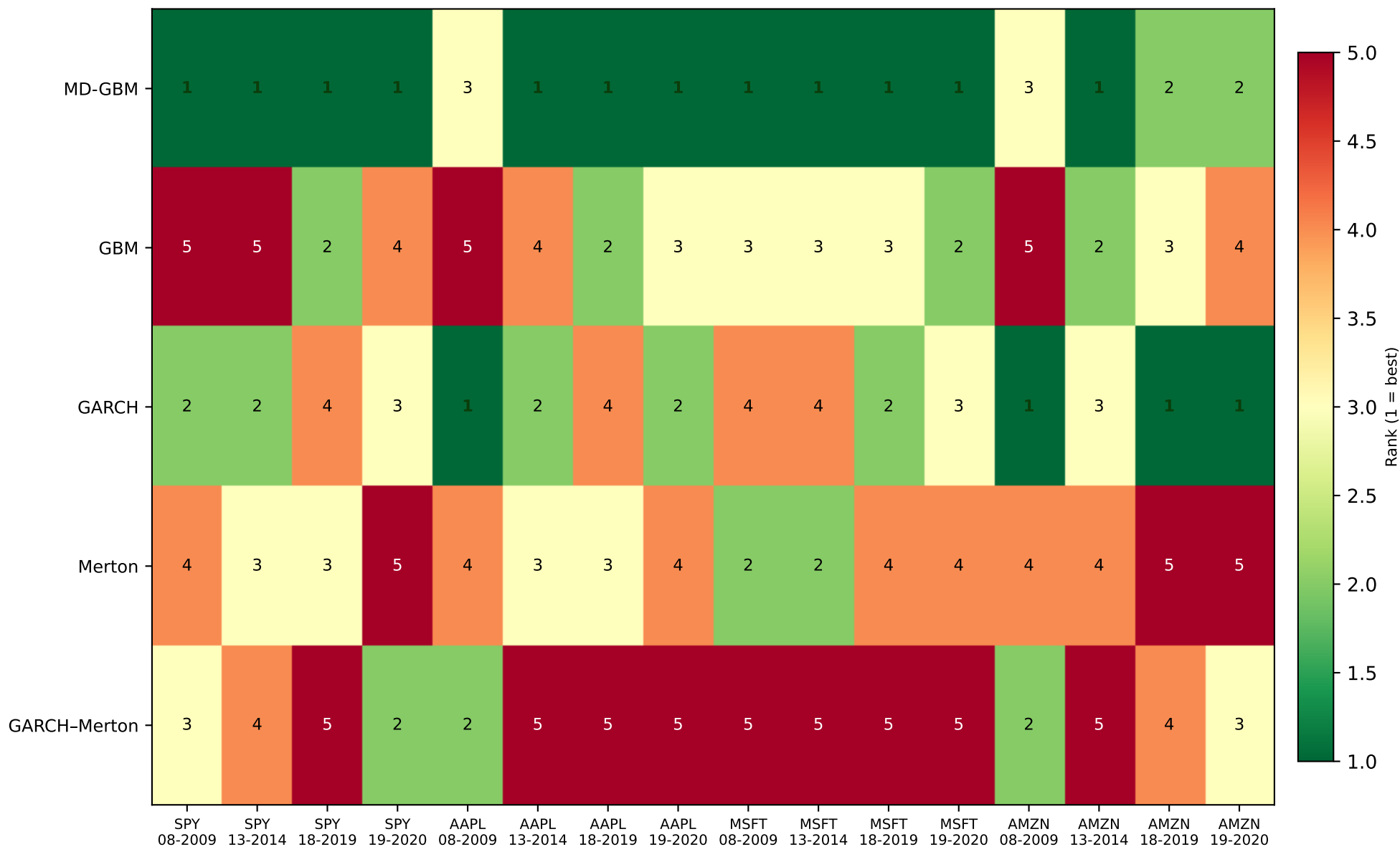
Model	Mean RMSE%	Median	# regimes best
MD-GBM	32.88	35.09	1
GBM	43.52	45.29	0
GARCH	31.20	31.28	3
Merton	47.73	47.93	0
GARCH-Merton	45.76	45.60	0

Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.

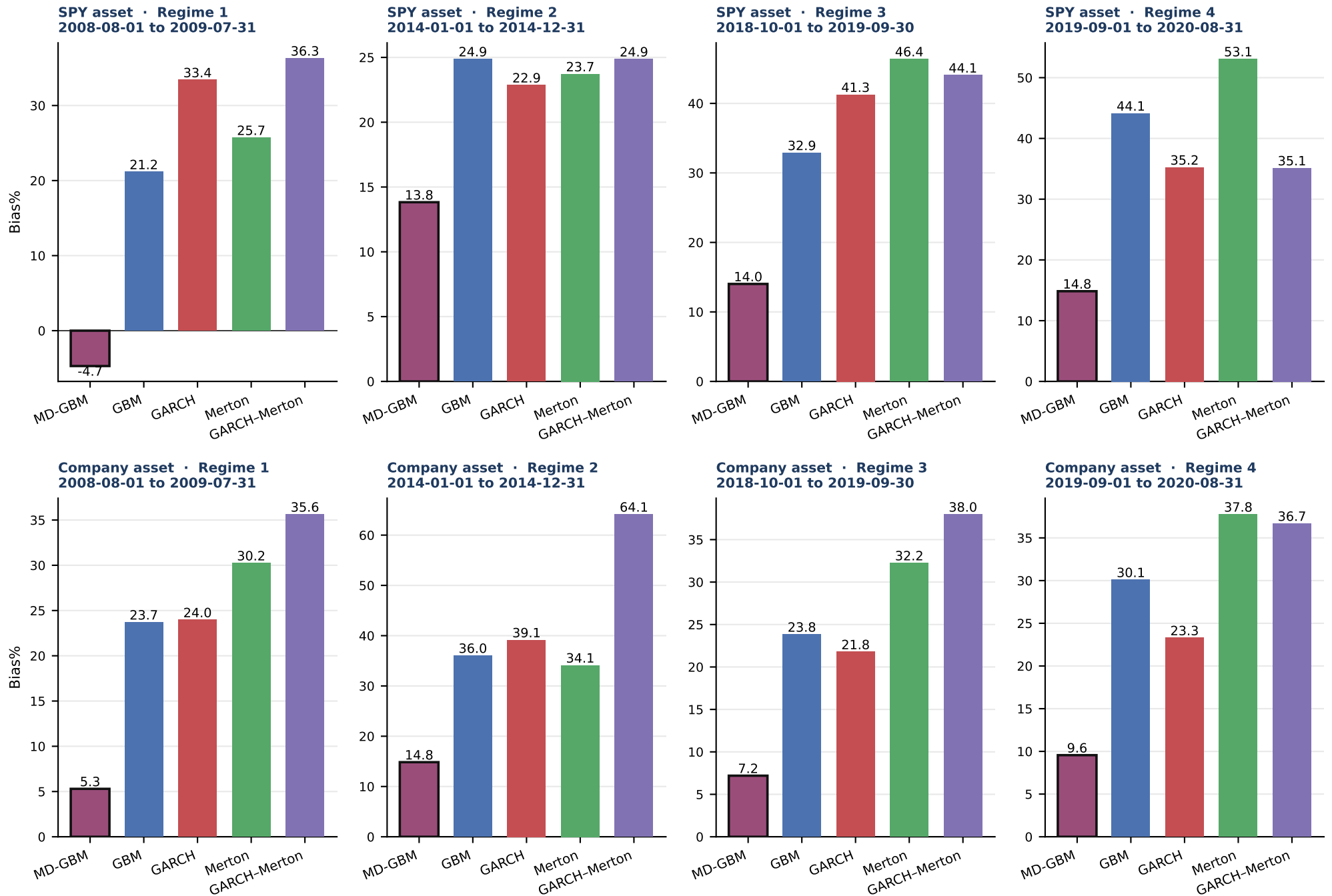




Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, MD-GBM is the best overall model (mean RMSE% = 33.94; 12 of 16 cells).
- Second overall is GARCH (mean RMSE% = 44.24; 4 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, MD-GBM.
- SPY: lowest mean RMSE% is MD-GBM (37.78); wins 4 of 4 regimes.
- AAPL: lowest mean RMSE% is MD-GBM (32.47); wins 3 of 4 regimes.
- MSFT: lowest mean RMSE% is MD-GBM (32.65); wins 4 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (31.20); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is MD-GBM (38.73).
- 2013-2014 (Normal): best mean RMSE% is MD-GBM (30.11).
- 2018-2019 (Late-cycle): best mean RMSE% is MD-GBM (30.08).
- 2019-2020 (COVID): best mean RMSE% is MD-GBM (36.85).
- No-jump models as a group have lower mean RMSE% (41.94) than the jump models in this report (51.93).
- For the overall winner (MD-GBM), crisis RMSE% is 38.73 vs 30.11 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all five models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE% and Bias% are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).